EXERCISE – 4.9

STANDARD NORMAL DISTRIBUTION

1. If $Z \sim N(0,1)$. Find

$(a)\ P(Z < 0.87)$ $(b)\ P(Z < 1.42)$

$(c)\ P(Z > 0.82)$ $(d)\ P(Z > 2.58)$

$(e)\ P(Z < -0.65)$ $(f)\ P(Z < -1.32)$

$(g)\ P(Z > -1.43)$ $(h)\ P(Z > -2.37)$

$(Ans: 0.8078,\ 0.9222,\ 0.2061,\ 0.0049,\ 0.2578,\ 0.0934,\ 0.9236,\ 0.9911)$

2. If $Z \sim N(0,1)$. Find

$(a)\ P(1.12 < Z < 1.43)$ $(b)\ P(1.46 < Z < 2.97)$

$(c)\ P(-0.98 < Z < -0.16)$ $(d)\ P(-1.76 < Z < -0.25)$

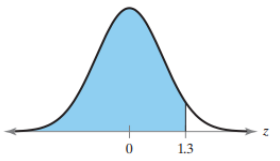
$(e)\ P(-0.20 < Z < 1.56)$ $(f)\ P(-2.46 < Z < 1.74)$

$(g)\ P(|Z| < 1.78)$ $(h)\ P(|Z| > 0.75)$

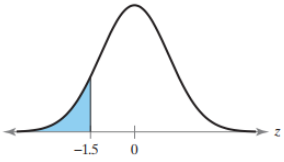
$(Ans: 0.0550,\ 0.0707,\ 0.2729,\ 0.3621,\ 0.5199,\ 0.9521,\ 0.9249,\ 0.4533)$

3. If $Z \sim N(0,1)$. Find the probabilities represented by the shaded areas

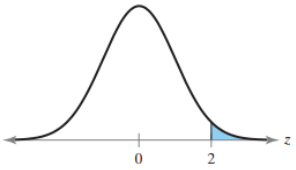
(a)



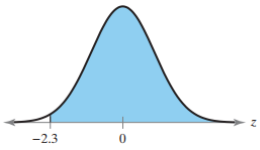
(b)



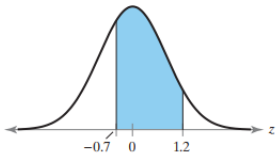
(c)



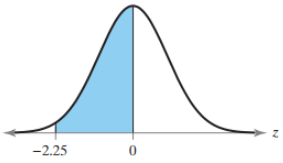
(d)



(e)



(f)



$(Ans: 0.9032,\ 0.0668,\ 0.0228,\ 0.9893,\ 0.6430,\ 0.4878)$

4. If $Z \sim N(0,1)$. Find

a. $P(0.00 < z < 0.74)$ b. $P(-1.17 < z < 1.94)$

c. $P(z > 1.25)$ d. $P(z < 1.75)$

$(Ans: 0.2703,\ 0.8528,\ 0.1056,\ 0.9599)$

5. If $Z \sim N(0,1)$. Find

a. $P(-3.05 < z < 0.00)$ b. $P(-2.43 < z < 1.37)$

c. $P(z < -2.17)$ d. $P(z > 2.43)$

$(Ans: 0.4989,\ 0.9071,\ 0.015,\ 0.0075)$

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GENERAL NORMAL DISTRIBUTION

6. If the random variable x is normally distributed with mean $\mu = 174$ and standard deviation $\sigma = 20$. Find the indicated probability

$(a)\ P(X < 180)$ $(b)\ P(X < 160)$

$(c)\ P(X > 185)$ $(d)\ P(X > 170)$

$(e)\ P(170 < X < 195)$ $(f)\ P(155 < X < 172)$

$(Ans: 0.8078,\ 0.9222, 0.2061,\ 0.0049,\ 0.2578, 0.0934)$

7. The random variable X is normally distributed such that $X \sim N(50, 20)$. Find

(a) $P(X > 60.3)$

(b) $P(X < 59.8)$

(c) $P(45 < X < 60)$

$(Ans: 0.0106, 0.9857, _____)$

8. The random variable X is normally distributed such that $X \sim N(-8, 12)$. Find

(a) $P(X < -9.8)$

(b) $P(X > -8.2)$

(c) $P(-7 < X < 0.5)$

$(Ans: 0.3015, 0.5231, 0.3792)$

9. The random variable X is normally distributed such that $X \sim N(100, 81)$. Find

(a) $P(|X - 100| < 18)$

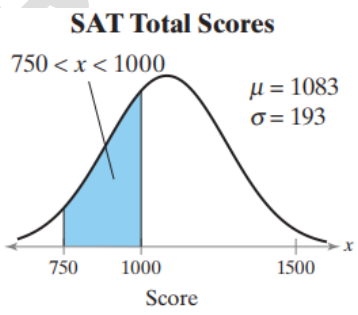
(b) $P(|X - 100| > 5)$

(c) $P(12 < X - 100 < 15)$

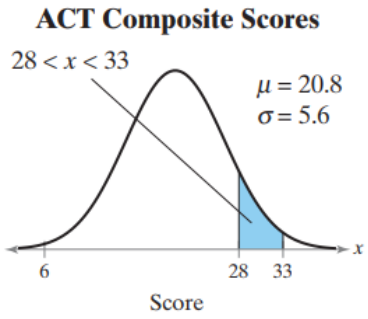
$(Ans: 0.9544, 0.5784, 0.0435)$

10. Assume the variable x is normally distributed. Find the probability of the shaded region of the graph.

(a)



(b)



$(Ans: 0.9544, 0.5784, 0.0435)$

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EXERCISE – 4.10

APPLICATIONS OF NORMAL DISTRIBUTION

1. The heights of boys at particular age follows a normal distribution with mean 150.3 cm and variance 25 cm. Find the probability that a boy picked at random from this age group
   - **a.** less than 153 cm,
   - **b.** more than 158 cm
   - **c.** between 150 cm and 158 cm.

   (*Ans* : 0.7054, 0.0618, 0.4621)

2. The masses of packages from a particular machine are normally distributed with a mean of 200 g and a standard deviation of 2 g. Find the probability that a randomly selected package from the machine weighs
   - **a.** less than 197 g,
   - **b.** more than 200.6 g
   - **c.** between 198.5 g and 199.5 g

   (*Ans* : 0.0668, 0.4013, 0.1747)

3. The average number of calories in a 1.5-ounce chocolate bar is 225. Suppose that the distribution of calories is approximately normal with s 10. Find the probability that a randomly selected chocolate bar will have
   - **a.** Less than 200 calories
   - **b.** more than 215 calories
   - **c.** Between 200 and 220 calories

   (*Ans* : 0.0062, ,0.3023)

4. The average credit card debt for college seniors is $3262. If the debt is normally distributed with a standard deviation of $1100, find these probabilities.
   - **a.** That the senior owes at least $1000
   - **b.** That the senior owes more than $4000
   - **c.** That the senior owes between $3000 and $4000

   (*Ans* : 0.9803, 0.2514, 0.3434)

5. The average monthly mortgage payment including principal and interest is $982 in the United States. If the standard deviation is approximately $180 and the mortgage payments are approximately normally distributed, find the probability that a randomly selected monthly payment is
   - **a.** More than $1000
   - **b.** More than $1475
   - **c.** Between $800 and $1150

   (*Ans* : 0.4602, 0.0031, 0.6676)

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6. According to the Federal Highway Administration's 2003 highway statistics the distribution of ages for licensed drivers has a mean of 44.5 years and a standard deviation of 17.1 years. Assuming the distribution of ages is normally distributed, what percentage of the drivers are:

- **a.** Younger than 25 years of age
- **b.** Older than 21 years of age
- **c.** Between the ages of 45 and 65

(*Ans* : 0.1271, 0.9153, 0.373)

7. According to the American College Test (ACT), results from the 2004 ACT testing found that students had a mean reading score of 21.3 with a standard deviation of 6.0. Assuming that the scores are normally distributed:

- **a.** Find the probability that a randomly selected student has a reading ACT score less than 20.
- **b.** Find the probability that a randomly selected student has a reading ACT score between 18 and 24.
- **c.** Find the probability that a randomly selected student has a reading ACT score greater than 30

(*Ans* : 0.4142, 0.3825, 0.0735)

8. In a survey of South African CEOs, the annual incomes were normally distributed, with a mean of $7.14 million and a standard deviation of $0.5 million. Find the probability that a randomly selected CEO has an annual income that is

- **a.** less than $6.5 million,
- **b.** more than 8 million.
- **c.** between $7 million and $7.5 million.

(*Ans* : 0.1003, 0.0427, 0.3745)

9. In a recent year, the MCAT scores for the critical analysis and reasoning skills portion of the test were normally distributed, with a mean of 124.9 and a standard deviation of 3.0. Find the probability that a randomly selected medical student who took the MCAT has a critical analysis and reasoning skills score that is

- **a.** less than 120,
- **b.** more than 130.
- **c.** between 122 and 128

(*Ans* : 0.0512, 0.0446, 0.6824)

10. The ages of prime ministers of Sri Lanka, when they were first sworn to office, are normally distributed with a mean of 58.21 years and a standard deviation of 11.56 years. Find the probability that the age of a randomly selected prime minister of Sri Lanka when he was sworn to office was

- **a.** less than 50 years,
- **b.** more than 70 years.
- **c.** between 55 and 60 years.

(*Ans* : 0.2388, 0.1539, 0.1709)

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